YZV406E Project Proposal Guidelines – 2026-2027 Fall Term

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For this term project, you are expected to propose a group project that addresses a specific problem in robotics and aims to develop a suitable solution. Projects may focus on robot simulation, data analysis, perception, decision-making, task planning, or safe robot behavior. Implementing the project on a physical robot is not mandatory, but it is possible where appropriate. Your proposal must be grounded in a well-cited primary reference paper published in one of the approved journals or conferences.

Suggested Project Topics:

- Robot Perception
- Object Affordances and Task Planning
- Robot Learning from Videos
- Robot Failure Detection and Recovery
- Human–Robot Interaction and Intent Recognition
- Multi-Robot Coordination and Task Allocation
- Vision-Language Models for Robot Task Understanding
- Vision-Language-Action (VLA) Models for Robotics

These topics are intended as starting points. You are encouraged to identify a specific problem within your chosen topic and develop a feasible project with a clearly defined methodology and experimental evaluation.

Proposal Structure and Required Components

Your project proposal must include the following components:

- **Title:** Provide a clear and concise title that reflects the research problem and the main idea of your proposed approach.
- **Problem Description and Primary Reference Paper:** Clearly define the problem you intend to address. Identify the primary reference paper selected from the approved list of journals and conferences, and explain its relevance to your research problem. Justify your choice of paper, including its significance and impact in the field.
- **Environment, Benchmark, or Dataset:** Specify the simulation/real-robot environment, experimental platform, benchmark suite, or dataset you plan to use for implementation and evaluation. Justify your choice and explain how it supports the investigation of your research problem.
- **Challenges and Limitations:** Critically analyze the primary reference paper and identify the key challenges, limitations, or unresolved research questions that motivate your project. Explain why addressing these issues is important.
- **Proposed Method and Novel Contributions:** Describe your proposed approach in sufficient technical detail. Explain how it differs from the original method, how it addresses the identified limitations, and what contributions it is expected to make. Clearly state the improvements you aim to achieve.
- **Research Questions and Evaluation Plan:** Formulate clear and specific research questions that your project aims to answer. Describe how you plan to evaluate your proposed approach, including the

baselines or alternative methods you will compare against, the evaluation metrics you will use, and the experiments you intend to conduct.

- **Team Roles and Individual Contributions:** Each student must take on distinct responsibilities within the team, and each team member's individual contributions must be clearly identified and documented in the project report.
- **References:** Provide complete bibliographic information for the primary reference paper and all other relevant publications and resources. Use a consistent academic citation style throughout.

The project should go beyond simply implementing an existing system. You are expected to systematically investigate a clearly defined problem using an appropriate method and a well-designed experimental evaluation. Ensure that your proposal is feasible, clearly scoped, and designed to enable systematic experimental evaluation.

Accepted List of Journals and Conferences

Please select your primary reference paper from the journals and conferences listed below. The paper must have been published within the last three years (2024–2026) and should be relevant to robot intelligence, robot learning, embodied AI, autonomous systems, or related areas of artificial intelligence.

Conferences:

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- Conference on Robot Learning (CoRL)
- Robotics: Science and Systems (RSS)
- Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- AAAI Conference on Artificial Intelligence (AAAI)
- International Joint Conference on Artificial Intelligence (IJCAI)
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- IEEE/CVF International Conference on Computer Vision (ICCV)
- ACM/IEEE International Conference on Human-Robot Interaction (HRI)

Journals

- IEEE Transactions on Robotics
- Science Robotics
- IEEE Robotics and Automation Letters
- The International Journal of Robotics Research (IJRR)
- Autonomous Robots
- Robotics and Autonomous Systems
- IEEE Transactions on Cognitive and Developmental Systems
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Image Processing
- Journal of Machine Learning Research
- International Journal of Computer Vision

- Pattern Recognition
- Pattern Recognition Letters
- Artificial Intelligence Review
- Autonomous Agents and Multi-Agent Systems
- International Journal of Social Robotics

Paper Selection Requirements

- Select a primary reference paper from one of the approved journals or conferences.
- The paper must have been published between 2024 and 2026, inclusive.
- The selected paper should be well-cited and demonstrate clear relevance to the research problem you intend to address.
- Your proposal must critically analyze the reference paper, identify its limitations or open research questions, and propose a potentially effective solution to address them
- Additional references may be drawn from other relevant publications, including older foundational works, when necessary to establish the research background or motivate the proposed approach.
- The primary reference paper must be an original research contribution rather than a survey, review, editorial, or tutorial article.
- The relevance and suitability of the selected paper will be evaluated based on its scientific contribution, its connection to robot intelligence, and its potential to motivate a feasible and well-defined research project.

Submissions

- The first draft submission is mandatory. All groups must submit an initial draft by the specified deadline. Failure to submit a proposal will result in **a VF grade**. Therefore, each group must submit at least a basic proposal outlining the core idea, the problem to be addressed, and the proposed approach.
- We may request revisions to your proposal during the first round, and you will be expected to incorporate the requested changes into your project proposal in the second round.
- The proposal must be a **maximum of 4 pages**, plus 1 additional page for references.
- **Only one member** of each group should submit the project proposal on behalf of the entire group.
- **Late submission will not be accepted.**

Important Dates:

- **First-Round Project Proposal Submission Deadline:** 10.11.2026
- **Project Presentations:** 30.12.2026

Note that Generative AI tools were used to assist in drafting and organizing this document, including the project proposal guidelines and the list of accepted journals and conferences. The instructor reviewed and finalized the content.